



Alex Chen

- He/him
- 34 years old
- Avid hiker
- High School Counselor

Personality: Agreeableness



Maya Sharma

- She/her
- 17 years old
- Guitar player
- High School Junior

Personality: Conscientiousness

Scenario: Maya failed in the math exam and asks advice from Alex.

Alex's goal: Make Maya feel safe from this failure.

Maya's goal: Find relief from her anxiety.

Perception F_{per}

Understand the partner's previous-turn video



Input: Perceived Alex video from turn $t - 1$

Turn t

Turn $t + 1$

Extracted Cues



Speech (T_s^t): 'Maya, I know you feel bad right now ...'



Emotion State (T_e^t): 'supportive ...'



Facial Expression (T_d^t): 'gentle; firm eyes ...'

Social Reasoning F_{rea}

Infer hidden mental status and choose the best response

Structured Observation (O_a^t)

Persona Profile (P_a)

Scenario & Social Goal (G_a)

Theory-of-mind Inference



Belief (β^t)

'Alex is not judging me ...'



Desire (δ^t)

'Alex wants to make me feel safe ...'



Intention (ι^t)

'Alex is trying to comfort me ...'



Emotion (ϵ^t)

'Alex seems supportive and concerned ...'



Utterance Candidates Generator (C^t)

Ensemble Selection

Empathy Check (U_{emp})

Strategic Check (U_{strat})

Coherence Check (U_{coh})

Best Utterance (D_a^t): 'Thanks Alex. I've done my best ...'

Feeling (E_a^t): I feel Relieved but still Anxious.

Turn t

Roles switch

Speaker (Maya)



Listener (Alex)

Expression F_{exp}

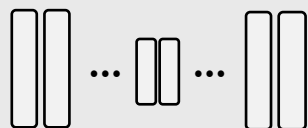
Render the selected response as a talking head video



Output: generated Maya video at turn t .

Turn t

Video Diffusion Model (G_{video})



portrait image (I_{ref}^a) reference audio (a_a^t)

